

Machine Learning

Unit-II (A): Logistic Regression

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Logistic Regression in Machine Learning

- Logistic Regression is a supervised machine learning algorithm used for classification problems.
- Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class.
- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs (real-valued number) into a probability value between 0 and 1.

Types of Logistic Regression

- Logistic regression can be classified into three main types based on the nature of the dependent variable:
 1. Binomial Logistic Regression
 2. Multinomial Logistic Regression
 3. Ordinal Logistic Regression

1. Binomial Logistic Regression

- This type is used when the dependent variable has only two possible categories.
- Examples include Yes/No, Pass/Fail or 0/1.
- It is the most common form of logistic regression and is used for binary classification problems.

2. Multinomial Logistic Regression

- This is used when the dependent variable has three or more possible categories that are not ordered.
- For example, classifying animals into categories like "cat," "dog" or "sheep."
- It extends the binary logistic regression to handle multiple classes.

3. Ordinal Logistic Regression

- This type applies when the dependent variable has three or more categories with a natural order or ranking.
- Examples include ratings like "low," "medium" and "high ".
- It takes the order of the categories into account when modelling.

Assumptions of Logistic Regression

To ensure the model is applied correctly, main assumptions are:

1. **Independent observations:** Each data point is assumed to be independent of the others means there should be no correlation or dependence between the input samples.
2. **Binary dependent variables:** It takes the assumption that the dependent variable must be binary, means it can take only two values. For more than two categories SoftMax functions are used.

Assumptions of Logistic Regression

3. **Linear relationship between independent variables and log odds:** The model assumes a linear relationship between the independent variables and the log odds of the dependent variable which means the predictors affect the log odds in a linear way.
4. **No outliers:** The dataset should not contain extreme outliers as they can distort the estimation of the logistic regression coefficients.
5. **Large sample size:** Logistic regression requires a sufficiently large sample size because it estimates parameters using maximum likelihood, and small samples lead to unstable and unreliable results

Note: Odds compare occurrence vs non-occurrence:

$$\text{Odds} = \frac{p}{1 - p}$$

Understanding Sigmoid Function

1. The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.
2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.
3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.
 - If the sigmoid output is same or above the threshold, the input is classified as Class 1.
 - If it is below the threshold, the input is classified as Class 0.

This approach helps to transform continuous input values into meaningful class predictions.

How does Logistic Regression work?

- Logistic regression models the probability of a binary outcome by applying the sigmoid (logistic) function to a linear combination of independent variables.
- The sigmoid function maps any real-valued input to a value between 0 and 1.

How does Logistic Regression work?

- Suppose we have input features represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ x_{21} & \dots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix}$$

- The dependent variable is Y having only binary value i.e., 0 or 1.

$$Y = \begin{cases} 0 & \text{if } Class1 \\ 1 & \text{if } Class2 \end{cases}$$

How does Logistic Regression work?

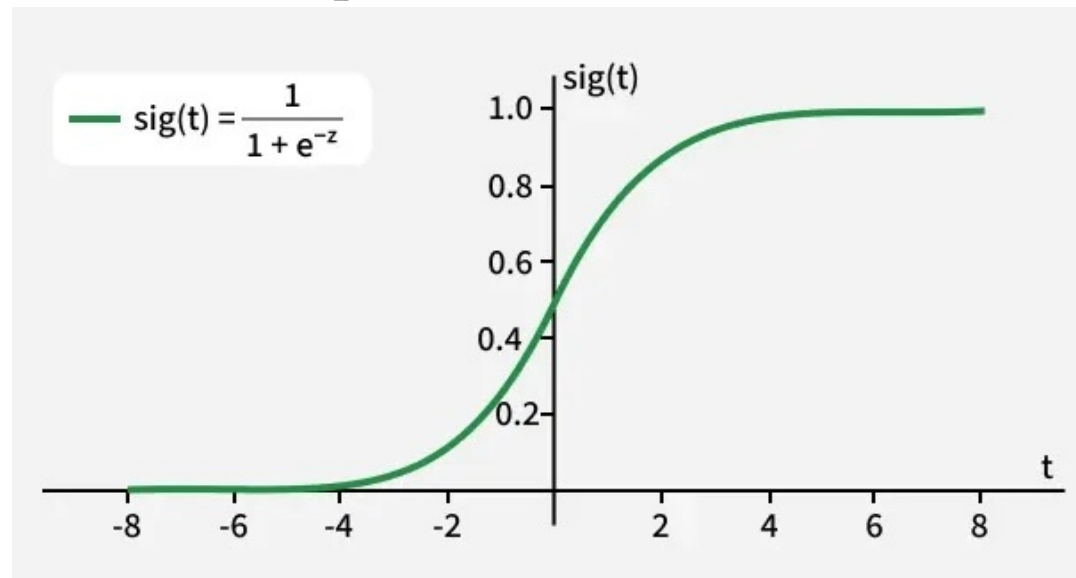
- Apply the multi-linear function to the input variables X .

$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$

- Here,
 - x_i is the i^{th} observation of X ,
 - $w_i = [w_1, w_2, w_3, \dots, w_m]$ is the weights or Coefficient,
 - b is the bias term also known as intercept.
- This can be represented as the dot product of weight and bias.
$$z = w \cdot X + b$$
- At this stage, z is a continuous value from the linear regression.

How does Logistic Regression work?

- Logistic regression then applies the sigmoid function to z to convert it into a probability between 0 and 1 which can be used to predict the class.
- Now we use the sigmoid function where the input will be z and we find the probability between 0 and 1, i.e., predicted y .
- $\sigma(z) = \frac{1}{1+e^{-z}}$



How does Logistic Regression work?

- The sigmoid function converts the continuous variable data into the probability, i.e., between 0 and 1.
- $\sigma(z)$ tends towards 1 as $z \rightarrow +\infty$ (Very Large Value)
- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$ (Very Small Value)
- $\sigma(z)$ is always bounded between 0 and 1
- where the probability of being a class can be measured as:
- $P(y = 1) = \sigma(z)$, It means probability that output belongs to class 1
- $P(y = 0) = 1 - \sigma(z)$, Probability that output belongs to class 0

Logistic Regression Equation and Odds

- It models the odds of the dependent event occurring which is the ratio of the probability of the event to the probability of it not occurring:
- Logistic regression assumes Log-odds is linear in X but probability is not linear in X, this distinction is critical.
- So, we are not deriving it from probability laws — we are choosing this form because it has good mathematical properties.
- $$\text{odds} = \frac{\text{Probability of event happening}}{\text{Probability of not happening}} = \frac{p(x)}{1-p(x)} = e^z$$
- Example:
- If $p = 0.8$; then Odds = $\frac{0.8}{0.2} = 4$
- Meaning event is 4 times more likely to occur than not occur.

Why Model Log-Odds?

- Probability $p(x)$ is between 0 and 1.
- Odds $\frac{p}{1-p}$ range from 0 to ∞ .
- Log-odds range from $-\infty$ to $+\infty$.
- And a linear model: $z = wX + b$, can produce any real number ($-\infty$ to $+\infty$).
- So log-odds perfectly match the range of a linear function.
- That's the key reason.
- Taking the natural logarithm of the odds gives the log-odds or logit:
- $\log \left[\frac{p(x)}{1-p(x)} \right] = z$
- $\log \left[\frac{p(x)}{1-p(x)} \right] = w \cdot X + b$

Example 1

For the customer service data, the proportion of customers who would recommend the service in the sample of customers is $p = 0.84$. Find the odds of recommending the service department.

Solution

The sample proportion of customers who would recommend the service is:

$$p = 0.84$$

Odds are calculated as:

$$\text{Odds} = \frac{p}{1 - p}$$

Substitute the value:

$$\text{Odds} = \frac{0.84}{1 - 0.84} = \frac{0.84}{0.16}$$

$$\text{Odds} = 5.25$$

Final Answer: The odds of recommending the service department are 5.25 to 1.

Question 1

If you deal one card from a standard deck, the probability that the card is a heart is $\frac{13}{52}$
 $= \frac{1}{4}$.

- a) Find the odds of drawing a heart.
- b) Find the odds of drawing a card that is not a heart.

Solution 1

Given:

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$$

a) Odds of drawing a heart

Odds are calculated as:

$$\text{Odds} = \frac{P(\text{event})}{1 - P(\text{event})}$$

$$\text{Odds of heart} = \frac{1/4}{1 - 1/4} = \frac{1/4}{3/4} = \frac{1}{3}$$

So, the odds of drawing a heart are 1 to 3.

Solution 1

b) Odds of drawing a card that is not a heart

$$P(\text{not heart}) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{Odds of not heart} = \frac{3/4}{1/4} = 3$$

So, the odds of drawing a card that is not a heart are 3 to 1.

Logistic Regression Equation and Odds

- $\frac{p(x)}{1-p(x)} = e^{w \cdot X + b}$
- $p(x) = e^{w \cdot X + b} \cdot (1 - p(x))$
- $p(x) = e^{w \cdot X + b} - e^{w \cdot X + b} \cdot p(x)$
- $p(x) + e^{w \cdot X + b} \cdot p(x) = e^{w \cdot X + b}$
- $p(x)(1 + e^{w \cdot X + b}) = e^{w \cdot X + b}$
- $p(x) = \frac{e^{w \cdot X + b}}{1 + e^{w \cdot X + b}}$

Logistic Regression Equation and Odds

- The final logistic regression equation will be:

$$p(X; \mathbf{b}, \mathbf{w}) = \frac{e^{\mathbf{w} \cdot \mathbf{X} + b}}{1 + e^{\mathbf{w} \cdot \mathbf{X} + b}} = \frac{1}{1 + e^{-\mathbf{w} \cdot \mathbf{X} + b}} = \frac{1}{1 + e^{-z}}$$

- This formula represents the probability of the input belonging to Class 1.

Question 1

The dataset of pass/fail in an exam for 5 students is given in the table below. If we use logistic regression as the classifier and assume the model suggested by the optimizer will become the following for Odds of passing a course:

$$\log(Odds) = -64 + 2 * Hours$$

Instance No.	Hours Studied	Result (Pass:1, Fail:0)
1	29	0
2	15	0
3	33	1
4	28	1
5	39	1

Question 1

1. Calculate the probability of pass for the student who studied 33 hours?
2. At least how many hours the student's study that makes sure will pass the course with the probability of more than 95% ?

Solution 1: Probability of passing for 33 hours

For $x = 33$:

$$z = -64 + 2(33) = -64 + 66 = 2$$

Now apply sigmoid:

$$p = \frac{1}{1 + e^{-2}}; \text{ } e^{-2} \approx 0.1353$$

$$p = \frac{1}{1 + 0.1353} = \frac{1}{1.1353} \approx 0.881$$

Probability of passing ≈ 0.881 (88.1%)

Solution 2: Hours required for probability $> 95\%$

We want:

$$p > 0.95$$

Using log-odds form:

$$\begin{aligned}\log\left(\frac{p}{1-p}\right) &> \log\left(\frac{0.95}{0.05}\right) \\ &= \log(19) \approx 2.944\end{aligned}$$

So:

$$\begin{aligned}-64 + 2x &> 2.944 \\ 2x &> 66.944 \\ x &> 33.472\end{aligned}$$

Students must study at least 34 hours to have probability $> 95\%$.

Likelihood Function for Logistic Regression

- The goal is to find weights w and bias b that maximize the likelihood of observing the data.
- For each data point i
 - for $y = 1$, predicted probabilities will be: $p(X; b, w) = p(x)$
 - for $y = 0$, predicted probabilities will be: $1 - p(X; b, w) = 1 - p(x)$
- Assuming data points are independent, the joint probability (likelihood) is the product of all individual probabilities:

$$L(b, w) = \prod_{i=1}^n p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}$$

- This is called the likelihood function.

Likelihood Function for Logistic Regression

- Taking natural logs on both sides:
 - $\log(L(b, w)) = \sum_{i=1}^n y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))$
 - $\log(L(b, w)) = \sum_{i=1}^n y_i \log p(x_i) + \log(1 - p(x_i)) - y_i \log(1 - p(x_i))$
 - $\log(L(b, w)) = \sum_{i=1}^n \log(1 - p(x_i)) + \sum_{i=1}^n y_i \log \frac{p(x_i)}{1 - p(x_i)}$
 - $\log(L(b, w)) = \sum_{i=1}^n \log\left(\frac{e^{-z_i}}{1 + e^{-z_i}}\right) + \sum_{i=1}^n y_i z_i$
 - $\log(L(b, w)) = \sum_{i=1}^n (-z_i) - (\log(1 + e^{-z_i})) + \sum_{i=1}^n y_i z_i$
 - $\log(L(b, w)) = \sum_{i=1}^n (-z_i) - (\log(1 + e^{z_i}) - z_i) + \sum_{i=1}^n y_i z_i$
 - $\log(L(b, w)) = \sum_{i=1}^n [y_i z_i - \log(1 + e^{z_i})]$
- This is known as the log-likelihood function.

Question 1

The dataset of pass/fail in an exam for 5 students is given in the table below. If we use logistic regression as the classifier and assume the model suggested by the optimizer will become the following for Odds of passing a course:

$$\log(Odds) = -64 + 2 * Hours$$

Instance No.	Hours Studied	Result (Pass:1, Fail:0)
1	29	0
2	15	0
3	33	1
4	28	1
5	39	1

Question 1

1. How to calculate the loss of model for the students who studied 33 hours?
2. What is total loss of the model ?

Solution 1: Loss of model for the students who studied 33 hours

Log-Likelihood Formula (Single Observation)

$$\ell_i = y_i z_i - \log(1 + e^{z_i})$$

For 33 hours: $y = 1, z = 2$

$$\ell = (1)(2) - \log(1 + e^2)$$

Compute:

$$e^2 = 7.389$$

$$\log(1 + 7.389) = \log(8.389) \approx 2.128$$

So:

$$\ell = 2 - 2.128 = -0.128$$

Loss (Negative Log-Likelihood)

$$\text{Loss} = -\ell = 0.128$$

Solution 2: Total loss of the model

Step 1: Compute z_i

Hours	y	$(z = -64 + 2x)$
15	0	-34
29	0	-6
28	1	-8
33	1	2
39	1	14

Solution 2: Total loss of the model

Step 2: Compute Individual Loss

1. $\mathbf{x} = 15, \mathbf{y} = 0$

$$L = -\log(1 + e^{z_i}) = -\log(1 + e^{-34}) \approx 0$$

2. $\mathbf{x} = 29, \mathbf{y} = 0$

$$L = -\log(1 + e^{-6}) = -\log(1 + 0.00247) = -0.00247$$

3. $\mathbf{x} = 28, \mathbf{y} = 1$

$$L = 1.(-8) - \log(1 + 0.00033) = -8 - 0.00033 = -8.00033$$

4. $\mathbf{x} = 33, \mathbf{y} = 1$

$$L = 1.(2) - \log(1 + 7.389) = -0.127$$

5. $\mathbf{x} = 39, \mathbf{y} = 1$

$$L = 1.(14) - \log(1 + 1202604.28) \approx 0$$

Solution 2: Total loss of the model

Step 3: Total Loss

$$\text{Total Loss} = 0 - 0.00247 - 8.00 - 0.127 - 0 = -8.130$$

$$\text{Average Loss } (-J) = -\left(\frac{-8.130}{5}\right) = 1.626$$

Conclusion

- The loss is large mainly because:
 - For 28 hours, the model predicted almost 0 probability,
 - But the actual label is 1,
 - A good logistic regression model typically has small average loss (close to 0).
 - Here $J = 1.626$ is quite large.
 - This shows the model does not fit the dataset well.

Question

A financial institution wants to predict whether a customer's credit card application will be approved (1) or rejected (0) based on the applicant's monthly income (in ₹ thousands).

- Suppose the logistic regression model learned by the optimizer gives the following equation for the log-odds of approval:
- $\log(\text{Odds}) = -15 + 0.6 \times \text{Income}$

Applicant No.	Monthly Income (₹ Thousands)	Approval Status (1 = Approved, 0 = Rejected)
1	25	0
2	18	0
3	35	1
4	30	1
5	40	1

Question

1. Calculate the probability that a customer with a monthly income of ₹28,000 will get the credit card approved.
2. What is the minimum monthly income required so that the probability of approval is greater than 90%?
3. Find the decision boundary income (where probability = 0.5).
4. Calculate the odds of approval for a customer earning ₹30,000.
5. Compute the log-loss for applicant 3.

Probability for Income = 28

Step 1: Compute log-odds

$$z = -15 + 0.6(28) = -15 + 16.8 = 1.8$$

Step 2: Convert to probability

$$p = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.8}}$$

Here, $e^{-1.8} \approx 0.165$

$$p = \frac{1}{1 + 0.165} \approx 0.858$$

Final Answer: $p \approx 85.8\%$

Minimum Income for Probability > 90%

We need:

$$p > 0.90$$

Step 1: Convert to log-odds

$$\frac{p}{1-p} = \frac{0.9}{0.1} = 9$$
$$\log(9) \approx 2.197$$

Step 2: Substitute into model

$$-15 + 0.6 \times \text{Income} > 2.197$$

$$0.6 \times \text{Income} > 17.197$$

$$\text{Income} > 28.66$$

Final Answer: Minimum Income ≈ 28.7 thousand

Decision Boundary ($p = 0.5$)

At $p = 0.5$:

$$\log\left(\frac{0.5}{0.5}\right) = 0$$

So,

$$-15 + 0.6 \times \text{Income} = 0$$

$$0.6 \times \text{Income} = 15$$

$$\text{Income} = 25$$

Decision Boundary: Income = 25 thousand

Odds for Income = 30

Step 1: Compute log-odds

$$z = -15 + 0.6(30) = -15 + 18 = 3$$

Odds:

$$\begin{aligned}\text{Odds} &= e^3 \\ e^3 &\approx 20.09\end{aligned}$$

Final Answer: Odds ≈ 20.09

Log-Loss for Applicant 3

Income = 35

Actual $y = 1$

Step 1: Compute probability

$$z = -15 + 0.6(35) - 15 + 21 = 6$$

$$p = \frac{1}{1 + e^{-6}}; \quad e^{-6} \approx 0.00248$$

$$p \approx 0.9975$$

Step 2: Compute Log-Loss

For $y = 1$:

$$\text{Loss} = -\log(p) = -\log(0.9975) \approx 0.0025$$

Final Answer: Log-loss ≈ 0.0025

Gradient of the log-likelihood function

- To find the best w and b we use gradient ascent on the log-likelihood function.

$$\log(L(b, w)) = \sum_{i=1}^n [y_i z_i - \log(1 + e^{z_i})]$$

- The gradient with respect to each weight w_j is:

$$\frac{\partial L}{\partial w_j} = \frac{\partial}{\partial w_j} \sum_{i=1}^n [y_i z_i - \log(1 + e^{z_i})]$$

- Differentiate term by term:

$$\sum_{i=1}^n \left[y_i \frac{\partial z_i}{\partial w_j} - \frac{\partial}{\partial w_j} \log(1 + e^{z_i}) \right]$$

Gradient of the log-likelihood function

- Derivative of z_i :
- $z_i = \sum_{j=1}^m w_j x_{ij} + b$
- $\frac{\partial z_i}{\partial w_j} = x_{ij}$
- So, $y_i \frac{\partial z_i}{\partial w_j} = y_i x_{ij}$
- Derivative of $\log(1 + e^{z_i})$:
- with respect to z : $\frac{\partial}{\partial z} \log(1 + e^z) = \frac{e^z}{1+e^z}$; take $(1 + e^z)$ as u , then $\frac{1}{u} \cdot \frac{\partial u}{\partial z}$ (Using chain rule)
- with respect to w_j : $\frac{\partial}{\partial w_j} \log(1 + e^{z_i}) = \frac{e^{z_i}}{1+e^{z_i}} \cdot \frac{\partial z_i}{\partial w_j} = \frac{e^{z_i}}{1+e^{z_i}} x_{ij}$ (Hint: If: $f = g(z)$ and $z = h(w)$ then $\frac{\partial f}{\partial w} = \frac{\partial g}{\partial z} \cdot \frac{\partial z}{\partial w}$)
- But: $\frac{e^{z_i}}{1+e^{z_i}} = \frac{1}{1+e^{-z_i}} = p(x_i)$
- So, $\frac{\partial}{\partial w_j} \log(1 + e^{z_i}) = p(x_i) x_{ij}$

Gradient of the log-likelihood function

- Combine Both Results:

- $\frac{\partial L}{\partial w_j} = \sum_{i=1}^n [y_i x_{ij} - p(x_i) x_{ij}]$

- $\frac{\partial L}{\partial w_j} = \sum_{i=1}^n (y_i - p(x_i)) x_{ij}$

- So, The gradient is:

(Actual output – Predicted probability) $\times$ Feature value

Gradient Ascent Update Rule

$$w_j^{new} = w_j^{old} + \alpha \sum_{i=1}^n (y_i - p(x_i)) x_{ij}$$

Example

A logistic regression model is used to predict whether a student will **pass (1)** or **fail (0)** based on the number of study hours.

Student	Study Hours (x)	Result (y)
1	1	0
2	2	0
3	3	1
4	4	1

Assume:

- Initial parameters: $w = 0$, $b = 0$
- Learning rate: $\eta = 0.1$
- Use **Gradient Ascent** to maximize the log-likelihood function.
- Perform **two iterations** of gradient ascent.
- Compute the updated values of w and b after each iteration.

Step 1: Compute Probabilities ($w = 0, b = 0$)

- So for all students:

$$p_i = 0.5$$

Step 2: Compute Gradients

Compute $y_i - p_i$

x	y	p	$y - p$
1	0	0.5	-0.5
2	0	0.5	-0.5
3	1	0.5	0.5
4	1	0.5	0.5

Step 2: Compute Gradients

Now:

- $\frac{\partial \ell}{\partial b} = (-0.5 - 0.5 + 0.5 + 0.5) = 0$
- $\frac{\partial \ell}{\partial w} = 1(-0.5) + 2(-0.5) + 3(0.5) + 4(0.5)$
- $\frac{\partial \ell}{\partial w} = -0.5 - 1 + 1.5 + 2 = 2$

Step 3: Update Parameters

After Iteration 1:

$$w = 0.2, b = 0$$

Iteration 2

Step 1: Compute z and p

x	$(z = 0 + 0.2x)$	$p = \frac{1}{1 + e^{-z}}$
1	0.2	0.5498
2	0.4	0.5987
3	0.6	0.6457
4	0.8	0.6899

Step 2: Compute Gradients

Compute $y - p$:

x	y	p	$y - p$
1	0	0.5498	-0.5498
2	0	0.5987	-0.5987
3	1	0.6457	0.3543
4	1	0.6899	0.3101

Step 2: Compute Gradients

Now,

- $\frac{\partial \ell}{\partial b} = -0.5498 - 0.5987 + 0.3543 + 0.3101 = -0.4841$
- $\frac{\partial \ell}{\partial w} = 1(-0.5498) + 2(-0.5987) + 3(0.3543) + 4(0.3101)$
- $\frac{\partial \ell}{\partial w} = -0.5498 - 1.1974 + 1.0629 + 1.2404 = 0.5561$

Final Parameters After Two Iterations

$$w = 0.2556 \text{ and } b = -0.0484$$

Question

A logistic regression model is used to predict whether a customer will **purchase a product (1)** or **not purchase (0)** based on the number of advertisements viewed.

Assume:

- Initial parameters: $w = 0, b = 0$
- Learning rate: $\eta = 0.1$
- Use **Gradient Ascent** to maximize the log-likelihood function.

Customer	Ads Viewed (x)	Purchase (y)
1	2	0
2	3	0
3	5	1
4	6	1

Tasks

1. Compute predicted probabilities for all observations at initialization.
2. Compute gradients:

$$\frac{\partial L}{\partial w}, \frac{\partial L}{\partial b}$$

3. Perform **first iteration** of gradient ascent and compute updated values of w and b .
4. Perform **second iteration** and compute updated values of w and b .

Step 1: Compute Probabilities ($\mathbf{w} = \mathbf{0}, \mathbf{b} = \mathbf{0}$)

$$p_i = \frac{1}{2} = 0.5, z_i = 0$$

For all points.

Step 2: Compute gradients

x	y	p	$y - p$	$(y - p)x$
2	0	0.5	-0.5	-1
3	0	0.5	-0.5	-1.5
5	1	0.5	0.5	2.5
6	1	0.5	0.5	3
			$Sum = 0$	$Sum = 3$

$$\frac{\partial L}{\partial b} = \sum_{i=1}^4 (y_i - p_i) = 0 \text{ and } \frac{\partial L}{\partial w} = \sum_{i=1}^4 (y_i - p_i)x_i = 3$$

Step 3: Update parameters

$$b_1 = 0 + 0.1(0) = 0 \text{ and } w_1 = 0 + 0.1(3) = 0.3$$

ITERATION 2

Now:

$$w = 0.3, b = 0$$

Step 1: Compute new probabilities ($z_i = 0.3x$)

x	z	p
2	0.6	0.646
3	0.9	0.711
5	1.5	0.818
6	1.8	0.858

Step 2: Compute Gradients

x	y	p	$y - p$	$(y - p)x$
2	0	0.646	-0.646	-1.292
3	0	0.711	-0.711	-2.133
5	1	0.818	0.182	0.91
6	1	0.858	0.142	0.852
			<i>Sum</i> = -1.033	<i>Sum</i> = -1.663

$$\frac{\partial L}{\partial b} = \sum_{i=1}^4 (y_i - p_i) = -1.033 \quad \text{and} \quad \frac{\partial L}{\partial w} = \sum_{i=1}^4 (y_i - p_i)x_i = -1.663$$

Step 3: Update again

$$w_2 = 0.3 + 0.1(-1.663) = 0.1337$$

$$b_2 = 0 + 0.1(-1.033)$$

$$b_2 = -0.103$$

Final Answers

After 1st iteration:

$$w_1 = 0.3, b_1 = 0$$

After 2nd iteration:

$$w_2 \approx 0.134, b_2 \approx -0.103$$

Terminologies involved in Logistic Regression

1. **Independent Variables:** These are the **input features or predictor variables** used to make predictions about the dependent variable.
2. **Dependent Variable:** This is the **target variable** that we aim to predict. In logistic regression, the **dependent variable is categorical**.
3. **Logistic Function:** This function **transforms the independent variables into a probability between 0 and 1** which represents the likelihood that the **dependent variable is either 0 or 1**.

Terminologies involved in Logistic Regression

4. **Odds:** This is the ratio of the probability of an event happening to the probability of it not happening. It differs from probability because probability is the ratio of occurrences to total possibilities.
5. **Log-Odds (Logit):** The natural logarithm of the odds. In logistic regression, the log-odds are modelled as a linear combination of the independent variables and the intercept.

Terminologies involved in Logistic Regression

6. **Coefficient:** These are the parameters estimated by the logistic regression model which shows **how strongly the independent variables affect the dependent variable.**
7. **Intercept:** The constant term in the logistic regression model which represents the **log-odds when all independent variables are equal to zero.**
8. **Maximum Likelihood Estimation (MLE):** This method is used to **estimate the coefficients of the logistic regression model by maximizing the likelihood of observing the given data.**

Linear v/s Logistic Regression

Aspect	Linear Regression	Logistic Regression
Definition	Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic regression is used to predict the categorical dependent variable using a given set of independent variables.
Problem Type	It is used for solving regression problem.	It is used for solving classification problems.
Output Type	In this we predict the value of continuous variables.	In this we predict values of categorical variables.
Curve/Model Fitting	In this we find best fit line.	In this we find S-Curve.

Linear v/s Logistic Regression

Aspect	Linear Regression	Logistic Regression
Estimation Method	Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
Output Example	The output must be continuous value such as price, age etc.	Output must be categorical value such as 0 or 1, Yes or No, etc.
Relationship Requirement	It required linear relationship between dependent and independent variables.	It not required linear relationship.
Collinearity	There may be collinearity between the independent variables.	There should be little to no collinearity between independent variables.

Thank You!